Polysaccharides

POLYSACCHARIDES

Definition:

- polysaccharides (greek: poly-many) are polymers of monosaccharide units with high molecular weight .
- Polysaccharides (or simply **glycans**) consist of repeat units of monosaccharides or their derivatives, held together by glycosidic bonds.
- They are usually tasteless (non-sugars) and form colloids with water.

Types of polysaccharides





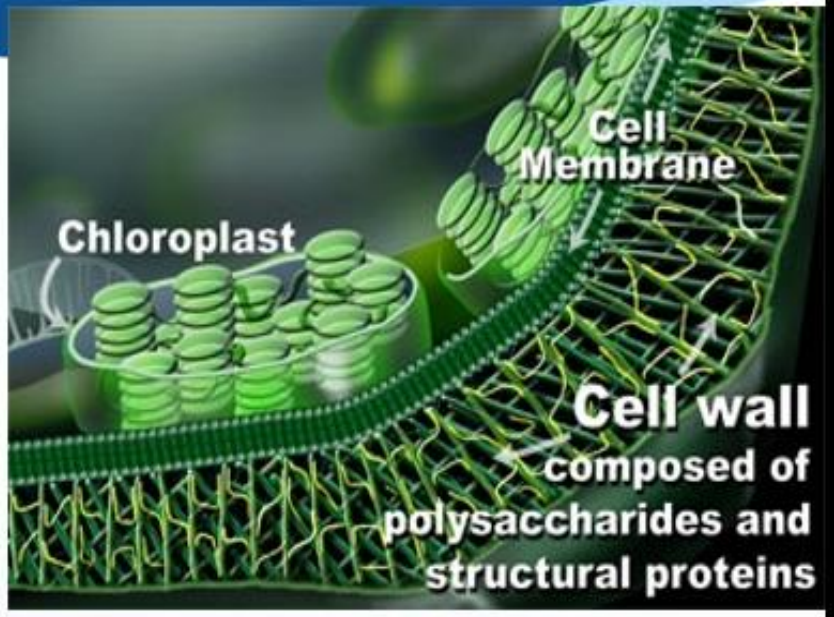
- **Starch**:
  - Used for energy storage in plants
  - Potatoes, pasta and rice are starches
  - They provide a quick form of energy for the body

I am formed in the Liver!

Glycogen:

- Used for energy storage in animals

More Polysaccharides



Cellulose:

- Provides structural support in plants (found in the cell wall)

Chitin:

- Found in exoskeletens of arthropods (insects, spiders)
- Found in cell wall of some fungi





Polysaccharides

Chemical:

- **Homopolysaccharides/Homoglycans- o**n hydrolysis yields a single type of monosaccharides.

- **Heteropolysaccharides/Heteroglycans-** on hydrolysis yields a mixture of different monosaccharides.

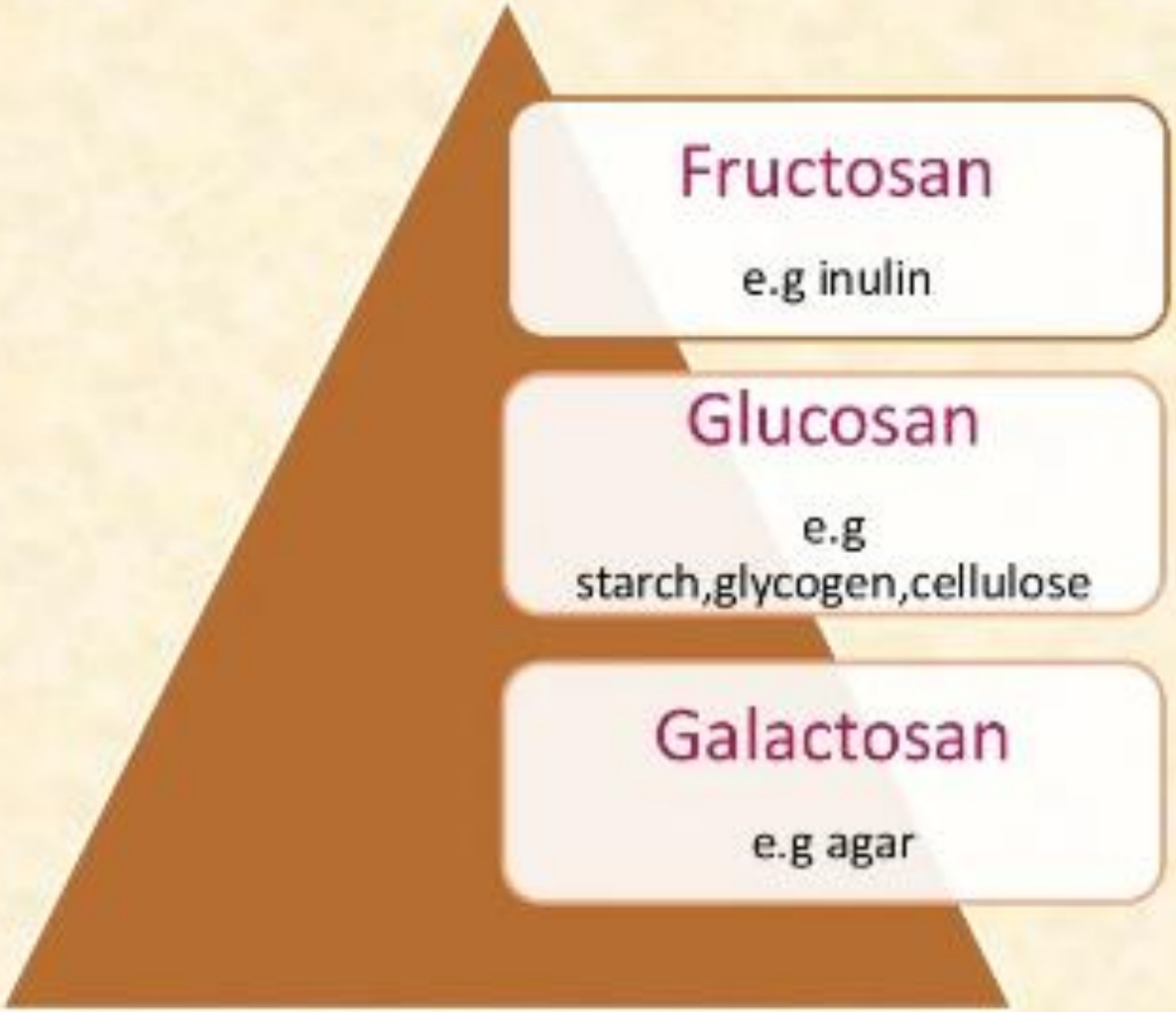
Functional:

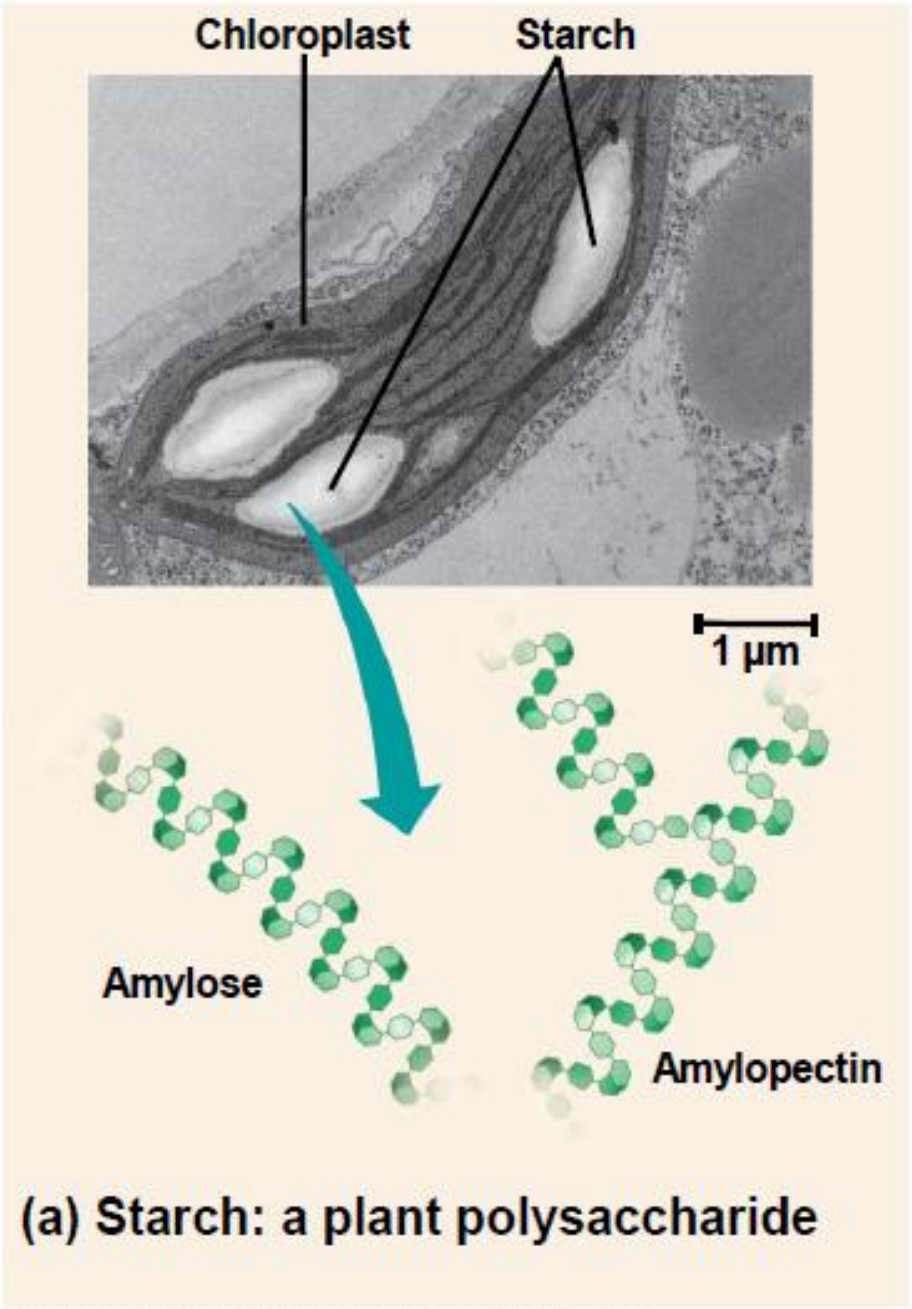
- **Nutrient polysaccharide (digestible)-** metabolic reserves of monosaccharides in plants and animals. **Eg. Starch, glycogen,**

- **Structural polysaccharide (indigestible)-**rigid mechanical structures in plants and animals. **Eg. Cellulose, pectin, chitin**

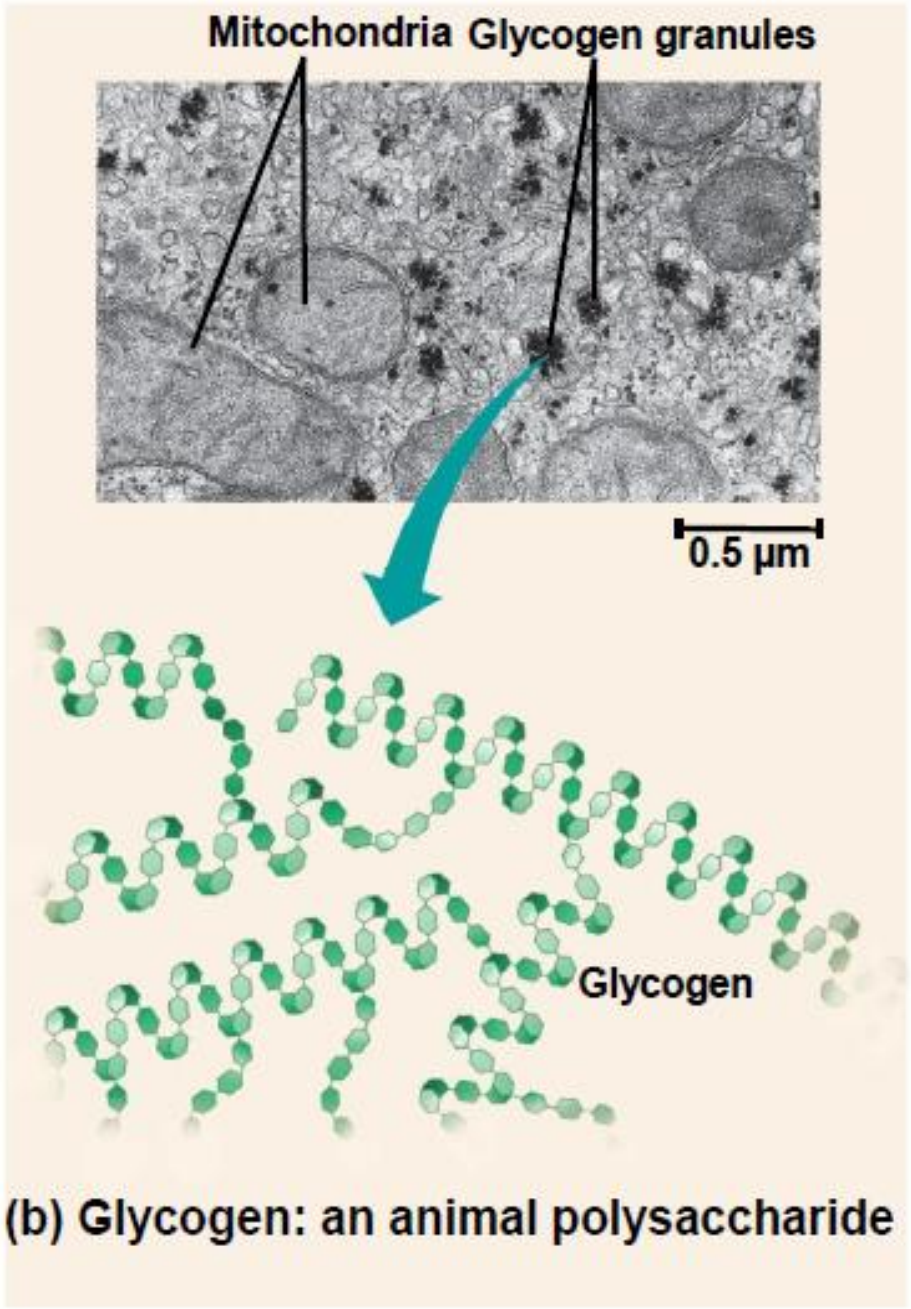
1.HOMOPOLYSACCHARIDES

Homopolysaccharides are polymers composed of single type of sugar units.





(a) Starch: a plant polysaccharide



(b) Glycogen: an animal polysaccharide

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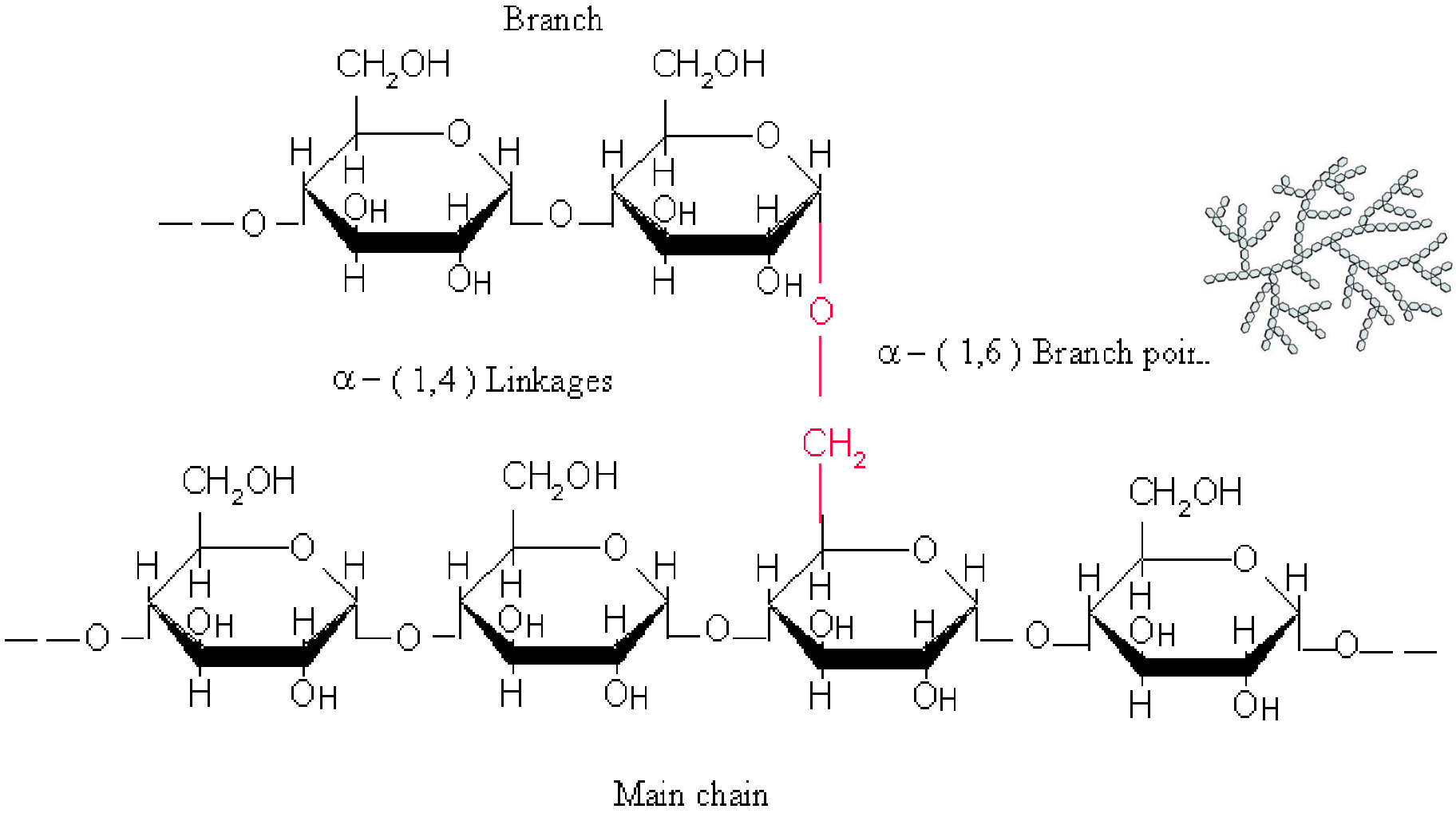
GLUCOSANS /GLUCANS

Glycogen (Storage Polysaccharide) –

- Also known as animal starch.
- Stored in muscle and liver.
- Present in cells as granules (high MW).
- Contains both α (1,4) links and α (1,6) branches at every 8 to 12 glucose unit.
- Complete hydrolysis yields glucose.
- With iodine gives a red-violet color.
- Hydrolyzed by both α and β-amylases

Glycogen

- **Alpha(1,6) branch point every 8-12 residues**



STARCH (STORAGE POLYSACCHARIDE)

- Most common storage polysaccharide in plants.

- Composed of 10 – 30% Amylose and 70-90% amylopectin depending on the source –

  (a) **Amylose** is a linear polymer of α-D-glucose, linked together by α -1→4 glycosidic linkages.

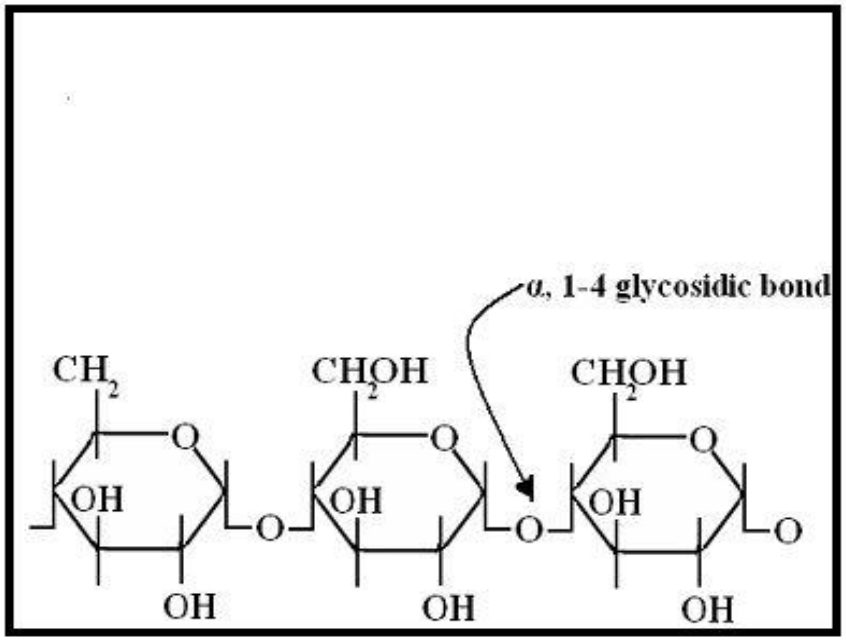
  It is soluble in water, reacts with iodine to give a blue color and the molecular weight of Amylose ranges between 50, 000 – 200, 000.

(b) **Amylopectin** is a highly branched polymer, insoluble in water, reacts with iodine to give a reddish violet color.

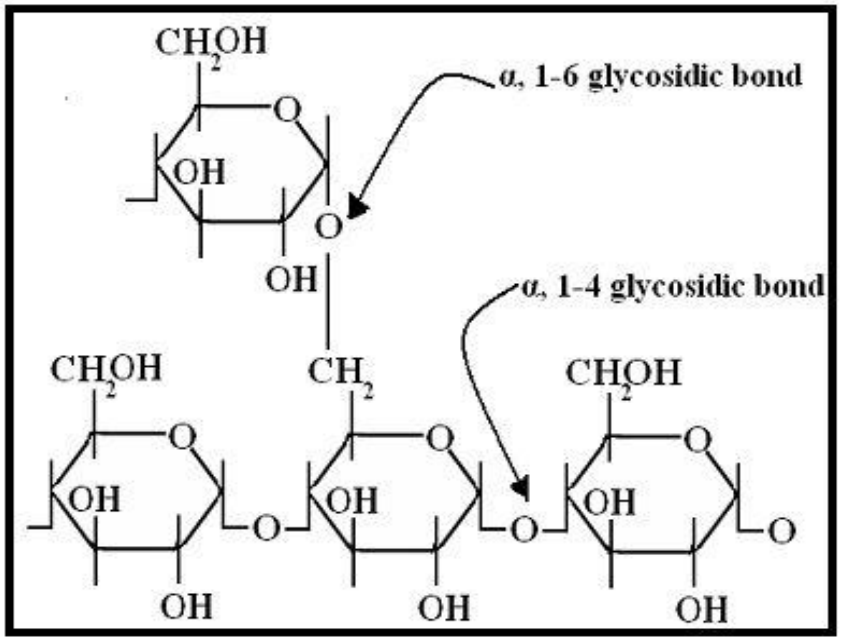
The molecular weight ranges between 70, 000 - 1 000, 000. Branches are composed of 25-30 glucose units linked by α-1→4 glycosidic linkage in the chain and by α-1→6 glycosidic linkage at the branch point.

Starch

- Alpha(1,6) branch point every 30 residues in amylopectin
- Most starch are 10-30% amylose and 70-90% amylopectin

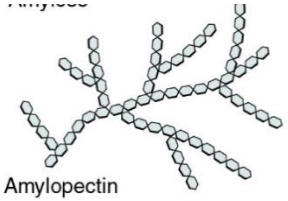


(a) amylose



(b) amylopectin





Amylose

1) Soluble in water

2)Gives Blue colour with dilute iodine solution

3)Structure –unbranched

4)250 to 300 D-glucose units linked by alpha 1→4 linkages

5)Mol. wt approx.60000

6)Occurs at the extent of 15 to 20%

Amylopectin

1)Insoluble in water

2)Gives reddish colour with iodine solution

3)Structure – highly branched

4)Units joined together by Alpha 1→4glycosidic bond and at branch point with alpha 1→6 glycosidic linkages

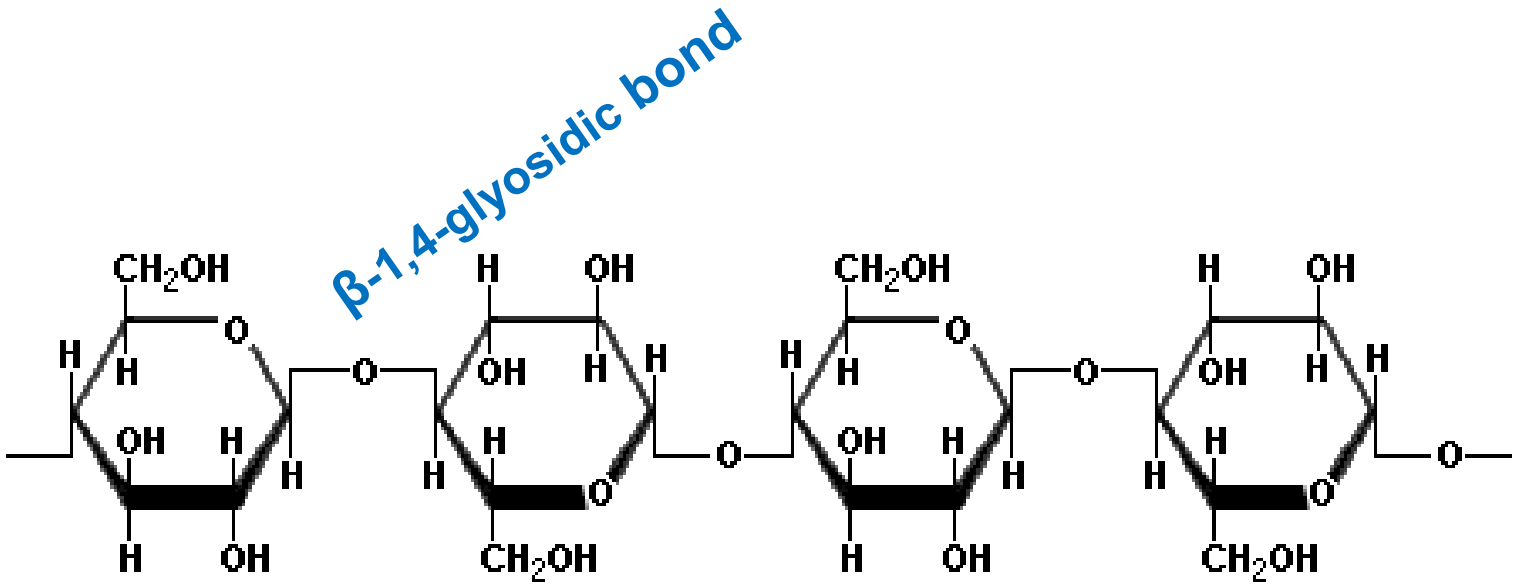
5)Mol. wt. approx 500000

6)0ccurs at 80 to 85%

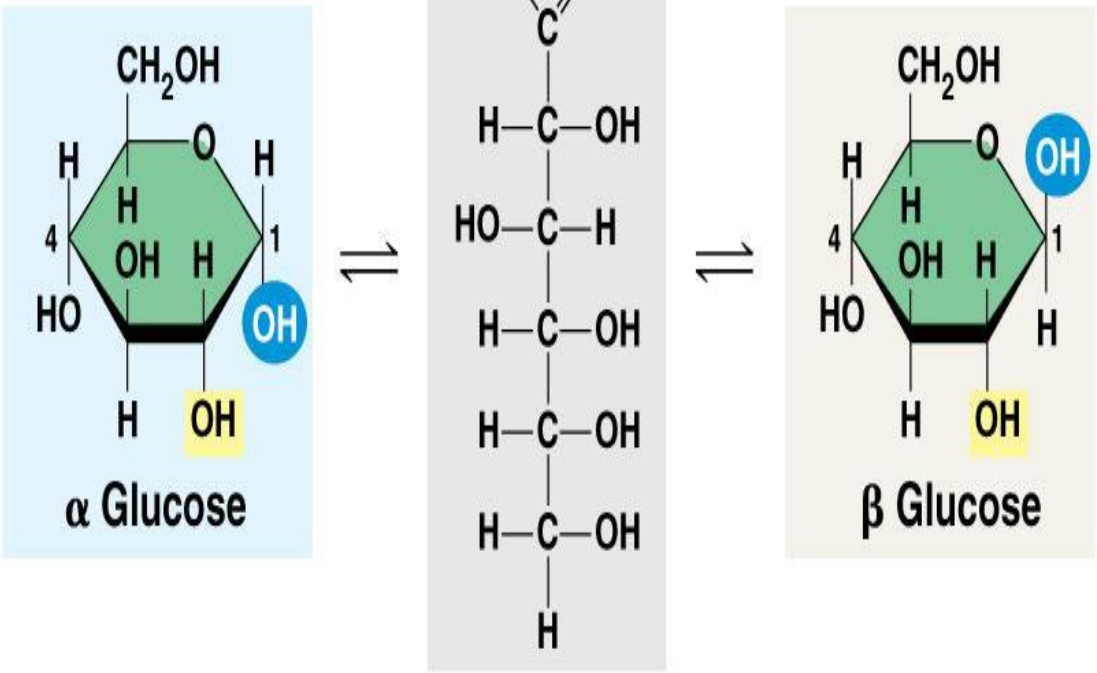
CELLULOSE (STRUCTURAL POLYSACCHARIDE)

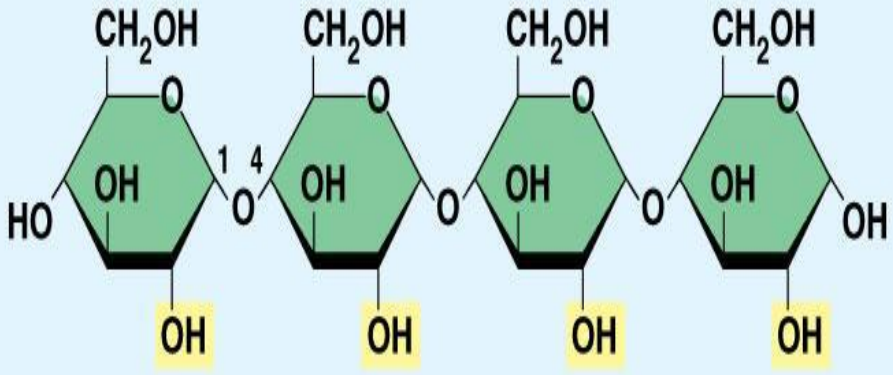
- Polymer of β-D-glucose linked by β(1,4) linkages.
- Yields glucose upon complete hydrolysis.
- Partial hydrolysis yields cellobiose.
- Most abundant of all carbohydrates.
- Gives no color with iodine.
- Cellulose is tasteless, odorless and insoluble in water and most organic solvents.

- Herbivorous animals utilise cellulose with the help of bacteria.
- Human beings lack any enzyme that hydrolyzes the β (1→ 4) bonds, and so cannot digest cellulose. It is an important source of "bulk" in the diet, and the major component of dietary fiber stimulating peristalsis and elimination of indigestible food residues.

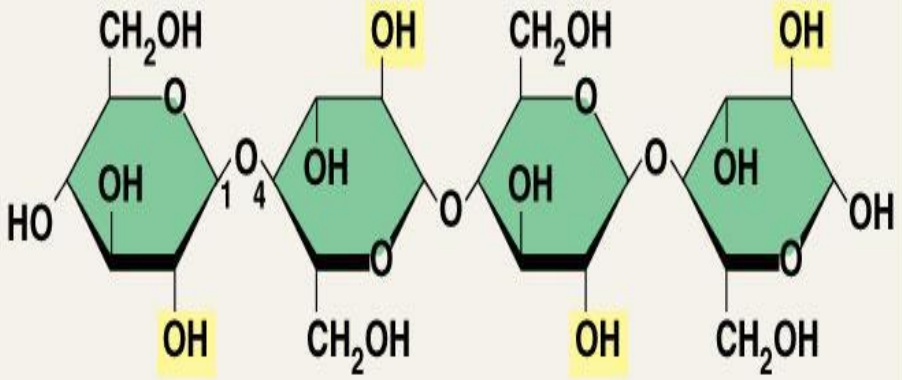


(a) α and β glucose ring structures



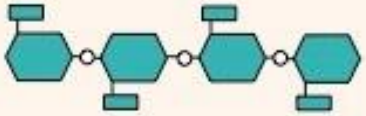
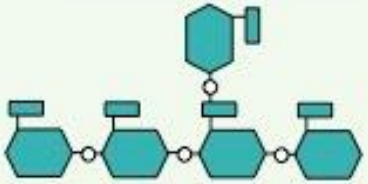
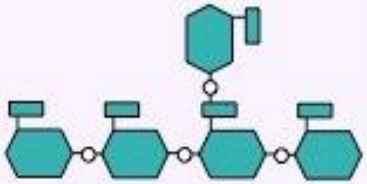
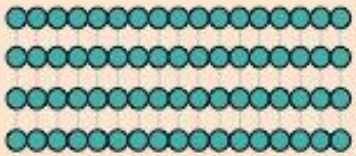
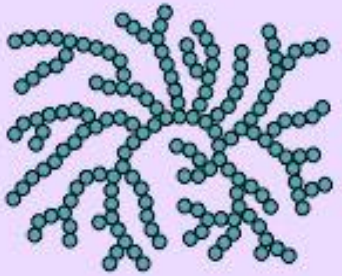


(b) Starch: 1–4 linkage of α glucose monomers



(c) Cellulose: 1–4 linkage of β glucose monomers

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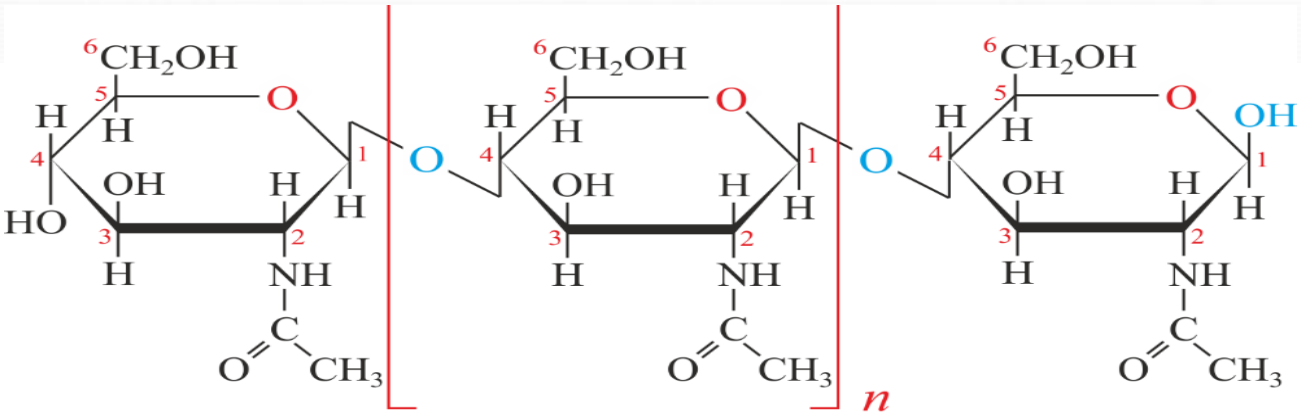
	Cellulose	Starch		Glycogen
		Amylose	Amylopectin	
Source	Plant	Plant	Plant	Animal
Subunit	β-glucose	α-glucose	α-glucose	α-glucose
Bonds	1-4	1-4	1-4 *and* 1-6	1-4 *and* 1-6
Branches	No	No	Yes (~per 20 subunits)	Yes (~per 10 subunits)
Diagram				
Shape				





Chitin

- chitin is the second most abundant carbohydrate polymer
- Like cellulose, chitin is a structural polymer
- present in the cell wall of fungi and in the exoskeletons of crustaceans, insects and spiders
- Consists of N-acetyl-D-glucosamine units joined by β(1→4) glycosidic linkages



β-(1,4)-N-acetyl-D-glucosamine

- **Polymer of N-acetyl-D-glucosamine with β-(1,4) glycosidic linkages**

Dextrans

- Products of the reaction of glucose and the enzyme transglucosidase from *Leuconostoc mesenteroides*.
- contains a (1,4), a (1,6) a (1,3) and a (1,2) linkages.
- MW: 4,000,000
- It is used in some eye drops as a lubricant.
- also used as molecular sieves to separate proteins and other large molecules (gel filtration chromatography)
- component of dental plaques.

FRUCTOSANS

Inulin



- β-(1,2) linked fructofuranoses
- linear only; no branching
- lower molecular weight than starch
- hydrolysis yields fructose
- sources include onions, garlic etc.
- used as diagnostic agent for the evaluation of glomerular filtration rate (renal function test)

GALACTOSANS

❖AGAR:-

- Agar is a galactose polymer.
- Obtained from the cell walls of some species of red algae (*Sphaerococcus Euchema*) and species of Gelidium.
- Dissolved in hot water and cooled, agar becomes gelatinous.
- Used in microbiology .

Heteropolysaccharides

The heteropolysaccharides are made up of more than one kind of monosaccharides and/or monosaccharide derivatives

The most important heteropolysaccharides in human beings are the mucopolysaccharides (glycosaminoglycans)

These are present in connective tissues and mucous secretions

Heteropolysaccharides

- Chemically, they are formed mostly of repeated disaccharides units that contain amino sugar (N-acetyl glucosamine or N-acetyl galactosamine) uronic acid (glucuronic acid or its 5 epimer iduronic acid).

3-types

1. **Glycosaminoglycans or mucopolysaccharides (GAGs)**
2. **Proteoglycans**
3. **Glycoproteins**

Glycosaminoglycans (GAGs)

- Glycosaminoglycans (GAGs) are large complexes of **negatively** charged **hetero**polysaccharide chains
- are associated with a small amount of protein, forming **proteoglycans**, which consist of over 95 percent carbohydrate
- bind with large amounts of water, producing the gel-like matrix that forms body's ground substance
- The viscous, lubricating properties of mucous secretions also result from GAGs, which led to the original naming of these compounds as mucopolysaccharides

Glycosaminoglycans (GAGs)

- **GAGs** are linear polymers of **repeating disaccharide** units

  [acidic sugar-amino sugar]n

- The amino sugar (usually sulfated) is either

  D-glucosamine or D-galactosamine

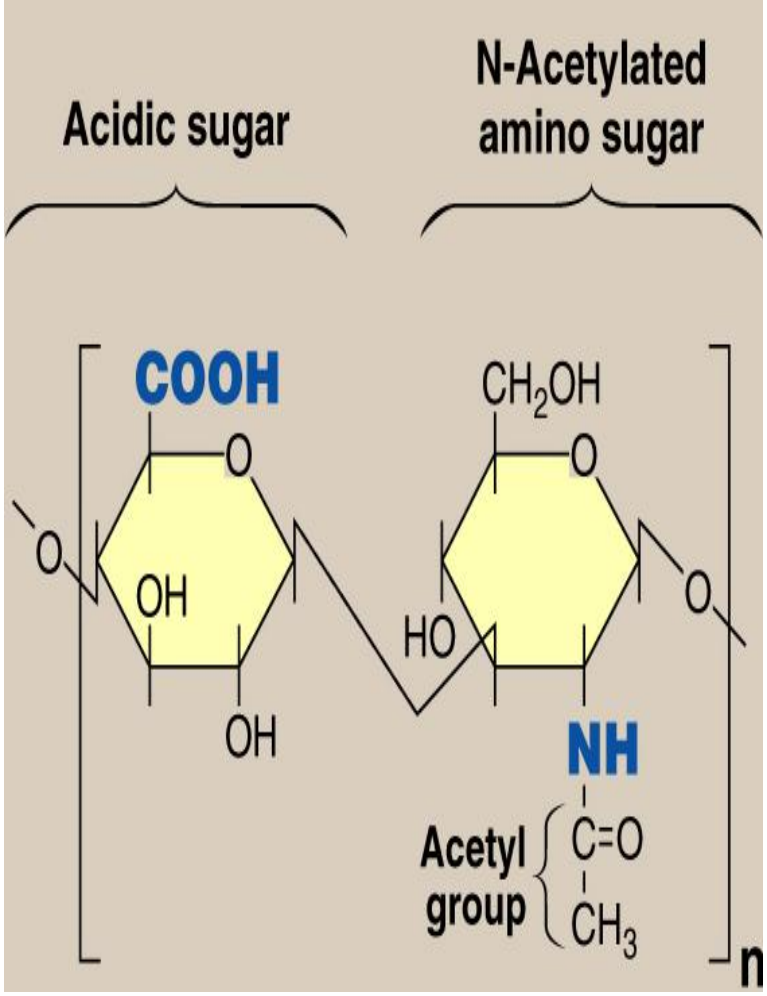
- The acidic sugar is either

  D-glucuronic acid or L-iduronic acid

- GAGs are strongly negatively-charged:

  carboxyl groups of acidic sugars

  Sulfate groups



Examples of GAGs

Examples of GAGs are:

1. **Chondroitin sulfates: Most abundant GAG**
2. **Keratan sulfates:** Most heterogeneous GAGs
3. **Hyaluronic acid: Compared to other GAGs, it is unsulfated and not covalently attached to protein**
4. **Heparin: Unlike other GAGs,** Unlike other GAGs that are extracellular, heparin is **intracellular and serves as an anticoagulant**

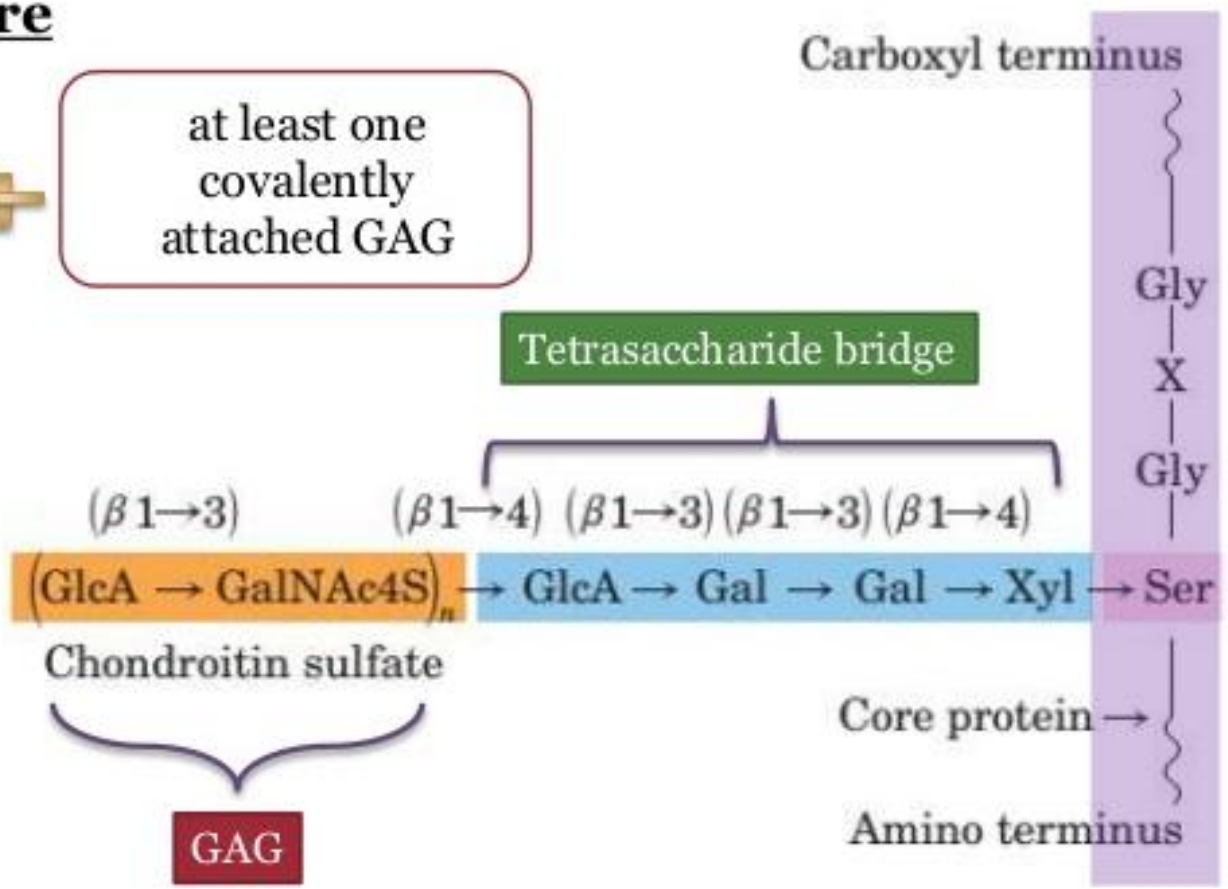
Proteoglycans

- Formed of GAGs covalently attached to core proteins.
- Carbohydrate content is 95% by weight
- **Occurrence:** found in all connective tissues, extracellular matrix (ECM) and on the surface of many cell types.
- Examples: **aggrecan, syndecan**

- **Functions:** role in glomerular filtration in kidney, act as receptors in cell membranes, in maintaining transparency of cornea.

Basic structure

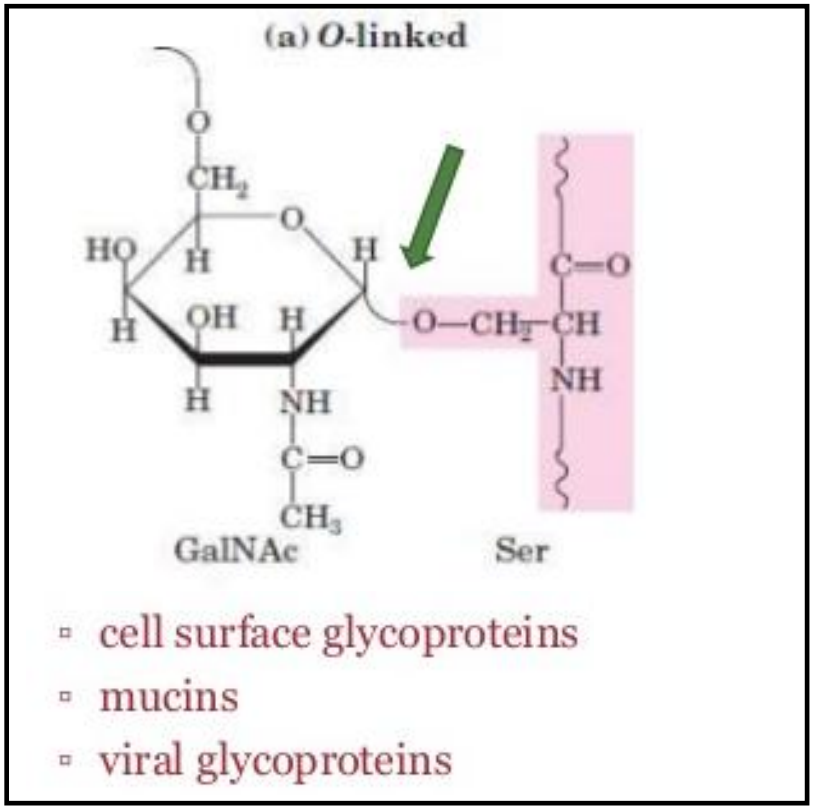
A core protein + at least one covalently attached GAG



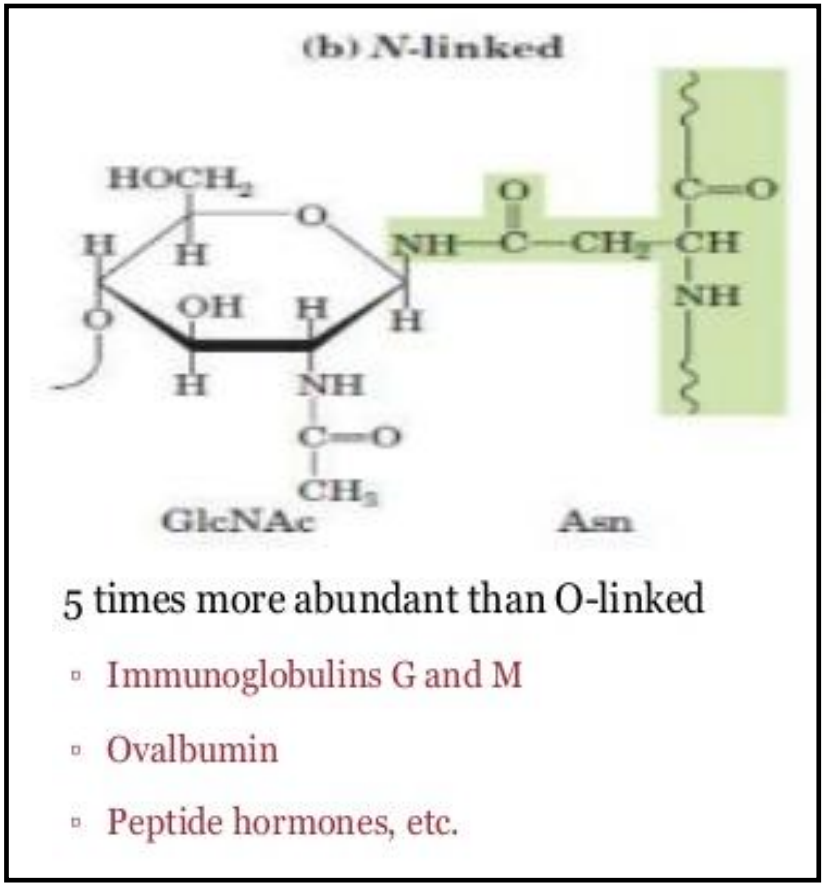
Lehninger principles of biochemistry: 5th ed.

GLYCOPROTEINS

- Covalently attached to proteins with much **smaller percentage of carbohydrate** than protein.
- Carbohydrates can be attached to the amide nitrogen in the side chain of asparagine **(N-linkage)** or to the hydroxyl oxygen of serine or threonine **(O-linkage)**



- cell surface glycoproteins
- mucins
- viral glycoproteins



5 times more abundant than O-linked

- Immunoglobulins G and M
- Ovalbumin
- Peptide hormones, etc.

SAMPLE QUESTIONS

1. A molecule of amylopectin which contains 1500 glucose residues and is branched after every 30 residues. How many reducing ends are there?

a) 0
b) 1
c) 2
d) 5

2. Humans are unable to digest

a) starch
b) complex carbohydrates
c) denatured proteins
d) cellulose

3. Which of the following is an analogous to starch?

a) Cellulose
b) Glycogen
c) Sucrose
d) Chitin

4. Non-digestible carbohydrates which serve as dietary fibres.

a) Glucose
b) Fructose
c) Cellulose
d) Maltose

5. Which of the following is an example of bacterial and yeast polysaccharide?

A. Starch
B. Glycogen
C. Cellulose
D. Dextrans

6. What is the chemical difference between cellulose and chitin?

A. Replacement of the hydroxyl group at C2 with an acetylated amino group
B. Replacement of the hydroxyl group at C3 with an acetylated amino group
C. Replacement of the hydroxyl group at C4 with an acetylated amino group
D. Replacement of the hydroxyl group at C5 with an acetylated amino group

7. When all the monosaccharides in a polysaccharide are same type, such type of a polysaccharide is called as _____

A. Glycogen
B. Homoglycan
C. Heteroglycan
D. Oligosaccharide

8. Which of the following are the storage polysaccharides?

A. Glycogen
B. Cellulose
C. Chitin
D. Glucose